

Import Data Using Transform Maps (Spreadsheet)

Project Report

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Name	Role
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1. INTRODUCTION

1.1 Project Overview

Many organizations keep their records (employees, assets, customers, incidents) in Excel spreadsheets. Moving this data into an enterprise platform by manual typing is slow and error-prone. This project uses the ServiceNow **Import Set** feature and **Transform Maps** to load spreadsheet data into a target table automatically. The spreadsheet is attached to a Data Source and loaded into a staging table; the Transform Map then maps each column to the correct field of the target table, checks for duplicates using a coalesce field, and inserts or updates the records.

1.2 Purpose

- Automate bulk data entry from spreadsheets into the platform.
- Reduce manual errors and save time.
- Prevent duplicate records through coalescing.
- Provide a reusable, repeatable import process with logs for tracking.
- Give the team hands-on experience with Import Sets, Data Sources, Field Maps and Transform Scripts.

2. IDEATION PHASE

2.1 Problem Statement

Data administrators receive data in spreadsheets and have to enter hundreds of rows by hand into the platform. This takes a long time, causes typing mistakes and duplicate entries, and offers no easy way to check what was loaded. A reliable method is needed to import spreadsheet data directly, map it to the right fields and validate it during the load.

Who does the problem affect?	What is the problem?	Why does it matter?	Impact
Data admins, IT support teams, business users	Manual entry of spreadsheet data into tables	Wrong or duplicate data reduces data quality	Time loss, errors, rework

2.2 Empathy Map Canvas

User: a data administrator who must load spreadsheet data into the platform.

SAYS "I have 500 rows to enter." "Can this be done faster?"	THINKS "Will I make mistakes?" "What if the same record is already there?"	DOES Opens Excel, copies rows, types them one by one into forms
FEELS Stressed by deadlines, bored by repetition, worried about errors	PAINS Time-consuming, duplicates, no validation, hard to track what failed	GAINS Fast bulk load, accurate mapping, import log, reusable process

2.3 Brainstorming

The team discussed several ways to bring spreadsheet data into the platform and compared them:

Idea	Suggested by (team)	Pros	Cons	Decision
Manual typing of each record	-	No setup	Slow, error-prone	Rejected
Custom script to read the file	Team discussion	Flexible	Needs heavy coding and maintenance	Rejected
Import Set + Transform Map (Excel)	Team discussion	Built-in, no-code mapping, logs, coalesce	Headings must match	Selected
Scheduled import via REST / SFTP	Team discussion	Fully automatic	More setup	Future scope

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User action	Thoughts / feelings	Opportunity
Awareness	Receives spreadsheet to load	Worried about effort	Introduce Import Sets
Preparation	Cleans headings and data in Excel	Careful, focused	Template for the sheet
Upload	Attaches file in Load Data	Hopeful	Simple upload wizard
Mapping	Reviews field maps, sets coalesce	Confident	Auto Map Matching Fields
Transform	Runs the transform map	Waiting for results	Clear import log
Verification	Checks target table and log	Relieved / fixes errors	Error report, re-run option

3.2 Solution Requirement

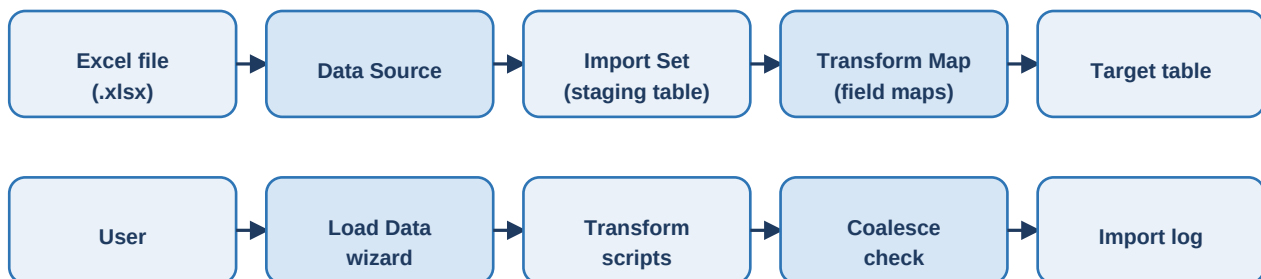
Functional requirements

ID	Requirement	Description
FR-1	Data source setup	Create a Data Source of type File / Excel with sheet number and header row
FR-2	Load data	Load spreadsheet rows into a staging (Import Set) table
FR-3	Field mapping	Map each staging column to a target-table field using Field Maps
FR-4	Coalesce	Update an existing record if the unique field matches, else insert a new one
FR-5	Validation	Use transform scripts to skip or fix invalid rows
FR-6	Logging	View inserted, updated, skipped and error counts in Transform History

Non-functional requirements

ID	Requirement	Description
NFR-1	Usability	Import can be done by a non-developer through the UI
NFR-2	Reliability	Same file can be re-run without creating duplicates
NFR-3	Performance	Hundreds of rows are processed within a short time
NFR-4	Security	Only users with import roles can load and transform data
NFR-5	Maintainability	Transform Map can be reused for future files with the same layout

3.3 Data Flow Diagram



Level-0 flow (top row): the spreadsheet is attached to a Data Source and loaded into the staging table; the Transform Map moves the data into the target table. Level-1 flow (bottom row): the user starts the load, scripts validate each row, coalesce decides insert or update, and the result is written to the import log.

3.4 Technology Stack

Layer	Technology
Platform	ServiceNow (Personal Developer Instance)
Data source	Microsoft Excel (.xlsx) file
Import mechanism	Import Sets, Data Source, Transform Maps, Field Maps
Scripting	Server-side JavaScript (onBefore / onAfter transform scripts)
Database	ServiceNow relational tables (staging and target)
Security	Roles (import_admin, import_transformer) and ACLs
Optional API	Import Set REST API / Table API

4. PROJECT DESIGN

4.1 Problem Solution Fit

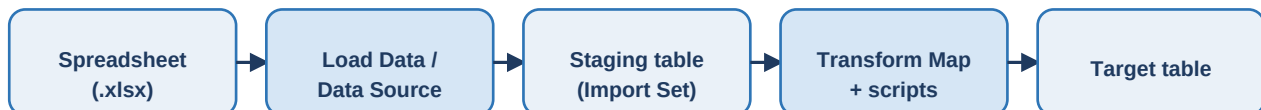
Problem	Solution fit
Manual entry is slow	Bulk import of all rows in one run
Duplicate records	Coalesce on unique field (e.g. emp_id)
Column names differ from table fields	Field Maps connect each source column to a target field
Hard to track errors	Import log and Transform History

Problem	Solution fit
Repeated imports needed	Reusable Data Source and Transform Map

4.2 Proposed Solution

Parameter	Description
Problem to be solved	Slow, error-prone manual loading of spreadsheet data
Idea / solution	Use Import Sets and Transform Maps to load Excel data into a target table
Novelty	Reusable no-code mapping with coalesce and script-based validation
Social / business impact	Saves staff time and improves data accuracy
Business model	Internal productivity tool for any ServiceNow-based organization
Scalability	Works for larger sheets, other tables, and can be scheduled later

4.3 Solution Architecture



Presentation layer: Load Data page, Transform Map forms, target-table list/form. **Logic layer:** Transform Map, Field Maps, coalesce and transform scripts. **Data layer:** staging table, target table and import-set log tables. **Security layer:** roles and ACLs.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Tasks	Owner	Status
Sprint 1 - Setup	Create PDI, target table, prepare sample spreadsheet	R Vinish, P Sri Ragavan	Completed
Sprint 2 - Import	Create Data Source, load data into staging table	R Vinish, E Dilli Kumar	Completed
Sprint 3 - Mapping	Create Transform Map, field maps, coalesce, scripts	K Jerina, E Dilli Kumar	Completed
Sprint 4 - Testing	Test valid/invalid/duplicate data, verify logs	B Mohanavalli	Completed
Sprint 5 - Documentation	Report, screenshots, demo	P Sri Ragavan, whole team	Completed

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test case	Input	Expected result	Result
Valid file	Rows with correct data	All rows inserted	Pass
Duplicate unique value	Same emp_id twice	Record updated, no duplicate	Pass
Missing mandatory field	Row with blank name	Row skipped and logged	Pass
Re-run same file	Same file loaded again	No duplicate records	Pass
Wrong heading	Renamed column	Field not mapped, detected at mapping	Pass
Larger file	Several hundred rows	Completes within a short time	Pass

7. RESULTS

7.1 Output Screenshots

Insert the following screenshots from your instance under these captions:

- Figure 1: Load Data page with the Excel file attached.
- Figure 2: Staging table with loaded rows.
- Figure 3: Transform Map and Field Map list.
- Figure 4: Transform result / Import Log (inserted, updated, errors).
- Figure 5: Target table showing the imported records.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Fast bulk loading of data No-code field mapping Avoids duplicates using coalesce Detailed logs for tracking Reusable for future files Script support for validation	Column headings must match exactly Needs clean, well-formatted spreadsheets Date/number format issues are possible Large files may need splitting Needs admin or import roles

9. CONCLUSION

The project shows that spreadsheet data can be imported into ServiceNow quickly and accurately using Import Sets and Transform Maps. The Data Source, staging table, field maps, coalesce and transform scripts together remove manual typing, prevent duplicates and give a clear log of results. The solution is simple to use, reusable and can be extended for larger and scheduled imports.

10. FUTURE SCOPE

- Scheduled imports from SFTP or a MID Server.
- Email notification with import summary.
- Automatic lookup of reference fields (department, location).
- A Service Portal page for easy file upload.

- Downloadable error report for failed rows.
- Support for CSV, XML and JSON sources.

11. APPENDIX

Source Code (transform script sample)

```
// onBefore transform script - skip rows without a name
(function runTransformScript(source, map, log, target) {
  if (!source.u_name || source.u_name.toString().trim() == '') {
    ignore = true;
    log.info('Row skipped: name is empty for emp_id ' + source.u_emp_id);
  }
  target.email = source.u_email.toString().toLowerCase();
})(source, map, log, target);
```

Dataset Link

Sample spreadsheet (employee_data.xlsx): [Insert link]

GitHub & Project Demo Link

GitHub: [Insert link]

Demo video: [Insert link]